CI/CD, Jenkins, Maven & Docker Interview Question Bank

Deduplicated, ranked by importance and grouped by topic
Prepared for a Test Automation Engineer with 4+ years of experience

Questions in your original document	**123**
Exact duplicates removed (same question, different wording of punctuation or case)	**1**
Near-duplicates merged (same question asked in different words)	**60**
Unique questions in this PDF	**62**

Section 1: Critical (Must Prepare)	18 questions
Section 2: High Priority	24 questions
Section 3: Medium Priority	16 questions
Section 4: Low Priority / Quick Revision	4 questions

How to Use This Document

Questions are ordered by how likely they are to be asked and how much weight interviewers give them for a tester with 4+ years of experience. Work top to bottom: finish Section 1 before you spend time on Section 3 or 4. Inside each section the questions are grouped by topic, and within each topic the most important question comes first.

A red tag such as **[asked x3 in your bank]** means the same question showed up several times in your original document (in different wording). Repetition is a good signal that it is a frequently asked question.

Section	What it means
Critical (Must Prepare) 18 questions	Asked in almost every automation interview at 4+ years. Describe how it works in YOUR project: which tool, which pipeline stages, what triggers runs and how failures are handled.
High Priority 24 questions	Frequent practical questions. Answer with the setup steps and one real example from your Jenkins/Maven/Docker work.
Medium Priority 16 questions	Occasionally asked, usually as follow-ups or when your resume mentions the tool.
Low Priority / Quick Revision 4 questions	Basic commands and rarely asked details. Skim them so you are never caught off guard.

What Interviewers Look For (4+ Years)

Topic	What the interviewer is testing
CI/CD Concepts & QA's Role	Can you explain the pipeline stages and exactly where your automation runs (smoke, regression, quality gates)?
Jenkins	Hands-on: freestyle vs pipeline, Jenkinsfile, parameters, scheduling with cron, triggering from Git, parallel jobs.
Reports, Failures & Troubleshooting	How tests are reported and what you do when a build fails or tests pass locally but fail in CI.
Maven & Build Management	Lifecycle, pom.xml, Surefire, dependencies and environment profiles.
Docker for QA	Running browsers and test environments in containers, Dockerfile and Compose basics.
Linux / CLI	Basic commands to inspect logs, processes and disk on a test server.

Note: short fragments in your file (for example 'Handling test failures' or 'Scheduling builds (cron)') are phrased as full questions and merged with matching ones. Docker and Linux commands are ranked lower because they are usually asked only as follow-ups.

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Topic Finder (all sections for one topic)

Use this if you want to study one topic at a time. Each number links to that topic's questions inside a section.

Topic	S1	S2	S3	S4	Total
CI/CD Concepts & QA's Role	6 Qs, p.5	5 Qs, p.6	4 Qs, p.8	-	**15**
Jenkins: Jobs, Pipelines, Triggers & Scheduling	6 Qs, p.5	5 Qs, p.6	2 Qs, p.8	1 Qs, p.9	**14**
Reports, Failures & Troubleshooting in CI	4 Qs, p.5	2 Qs, p.6	-	-	**6**
Maven & Build Management	2 Qs, p.5	3 Qs, p.6	-	-	**5**
Docker for QA Automation	-	6 Qs, p.6	8 Qs, p.8	1 Qs, p.9	**15**
Linux / CLI Basics for QA	-	3 Qs, p.7	2 Qs, p.8	2 Qs, p.9	**7**

S1 = Critical, S2 = High, S3 = Medium, S4 = Low priority.

Section 1: Critical (Must Prepare) (18 questions)

Asked in almost every automation interview at 4+ years. Describe how it works in YOUR project: which tool, which pipeline stages, what triggers runs and how failures are handled.

CI/CD Concepts & QA's Role (6)

1. What is CI/CD and why is it important in automation and for QA engineers? **[asked x3 in your bank]**
2. What CI/CD tools have you used? What tools can be used for CI/CD in QA automation? **[asked x2 in your bank]**
3. Explain the CI/CD pipeline and its common stages from a QA perspective. **[asked x2 in your bank]**
4. What is the role of QA in a CI/CD pipeline, and how does CI/CD improve software quality? **[asked x2 in your bank]**
5. How do you integrate automated tests, scripts and your framework into a CI/CD pipeline? Is your framework CI/CD ready? **[asked x5 in your bank]**
6. How do you integrate your Selenium suite with Jenkins, GitHub Actions or Azure DevOps? **[asked x3 in your bank]**

Jenkins: Jobs, Pipelines, Triggers & Scheduling (6)

7. What is Jenkins and why is it important for QA? How and when do you use it in your automation? **[asked x4 in your bank]**
8. What is the difference between a Freestyle job and a Pipeline job? Explain the basics of a Jenkins pipeline. **[asked x4 in your bank]**
9. What is a Jenkinsfile and why does QA use it? **[asked x3 in your bank]**
10. How do you create and configure a Jenkins job to run (trigger) your automation suite? **[asked x5 in your bank]**
11. How do you parameterize a Jenkins build / pass parameters to a job? **[asked x2 in your bank]**
12. How do you schedule a nightly regression run in Jenkins (cron)? **[asked x2 in your bank]**

Reports, Failures & Troubleshooting in CI (4)

13. How are automation reports published and stored in Jenkins? How do you publish HTML and TestNG reports and share them? **[asked x5 in your bank]**
14. How do you handle failed builds and test failures in Jenkins? **[asked x3 in your bank]**
15. Tests pass locally but fail on Jenkins/CI nodes. How do you debug? **[asked x2 in your bank]**
16. How do you handle flaky or unstable tests in CI/CD pipelines? **[asked x2 in your bank]**

Maven & Build Management (2)

17. What is Maven and why do you use it in automation projects? What is pom.xml and which basic Maven commands do you use? **[asked x3 in your bank]**
18. What is the Maven build lifecycle? **[asked x2 in your bank]**

Section 2: High Priority (24 questions)

Frequent practical questions. Answer with the setup steps and one real example from your Jenkins/Maven/Docker work.

CI/CD Concepts & QA's Role (5)

19. What is Continuous Integration (CI) and what is Continuous Delivery (CD)? **[asked x2 in your bank]**
20. What is the difference between Continuous Delivery and Continuous Deployment?
21. How do you trigger a CI/CD pipeline automatically after code changes (Git triggers)? **[asked x3 in your bank]**
22. What are quality gates in CI/CD and how does QA use them?
23. How do you manage test data in CI/CD?

Jenkins: Jobs, Pipelines, Triggers & Scheduling (5)

24. How do you run tests in parallel in Jenkins (parallel jobs)? **[asked x3 in your bank]**
25. How do you run a Maven project in Jenkins?
26. How do you integrate Postman API tests in Jenkins?
27. How do you use Jenkins to test across multiple environments?
28. Which Jenkins plugins have you used in test automation?

Reports, Failures & Troubleshooting in CI (2)

29. How do you archive test artifacts in Jenkins? **[asked x2 in your bank]**
30. How do you set up email and Slack notifications after builds? **[asked x2 in your bank]**

Maven & Build Management (3)

31. What is the Maven Surefire plugin and where and why do you use it?
32. How do you manage different environments (QA, pre-production, production) with different data using Maven / pom.xml? **[asked x2 in your bank]**
33. How are dependencies handled in Maven at run time? If someone deleted all dependencies from the pom file, can you still execute tests? **[asked x2 in your bank]**

Docker for QA Automation (6)

34. What is Docker and why is it useful in QA testing? How is it different from virtual machines? **[asked x2 in your bank]**
35. What is the difference between a Docker image and a container?
36. How do you write a Dockerfile and run a Docker container for automation testing (including API testing)? **[asked x3 in your bank]**
37. How do you run Selenium in Docker and use Docker for cross-browser testing? **[asked x2 in your bank]**
38. How do you integrate Docker with Jenkins for QA automation?
39. How do you share test environments using Docker Compose?

Linux / CLI Basics for QA (3)

40. What is the CLI and why should QA engineers know it?
41. How do you search inside a file for a keyword (grep)?
42. How do you monitor logs in real time (tail -f)?

Section 3: Medium Priority (16 questions)

Occasionally asked, usually as follow-ups or when your resume mentions the tool.

CI/CD Concepts & QA's Role (4)

43. What is a blue-green deployment and how does it help QA?
44. What is the difference between CI/CD for backend APIs and for frontend UI testing?
45. How do you test a CI/CD pipeline, its rollback mechanism and its scheduling? **[asked x3 in your bank]**
46. How do you schedule deployments?

Jenkins: Jobs, Pipelines, Triggers & Scheduling (2)

47. What are shared libraries in Jenkins?
48. How do you secure Jenkins for QA projects?

Docker for QA Automation (8)

49. What are common Docker commands every QA should know?
50. What is a Docker Volume and how do you persist test results in containers? **[asked x2 in your bank]**
51. How do you handle environment variables in Docker for QA tests?
52. How do you debug test failures in Docker containers?
53. How do you run database containers for test automation?
54. What is the difference between docker run and docker exec?
55. How do you optimize Docker images for faster QA pipeline execution?
56. What is Docker Hub and how do QA engineers use it?

Linux / CLI Basics for QA (2)

57. How do you see running processes and check disk and memory usage? **[asked x3 in your bank]**
58. How do you find a file in Linux and check network connectivity? **[asked x2 in your bank]**

Section 4: Low Priority / Quick Revision (4 questions)

Basic commands and rarely asked details. Skim them so you are never caught off guard.

Jenkins: Jobs, Pipelines, Triggers & Scheduling (1)

59. What are the two components Jenkins is integrated with?

Docker for QA Automation (1)

60. What is the difference between docker ps and docker images? How do you remove unused containers and images? **[asked x2 in your bank]**

Linux / CLI Basics for QA (2)

61. How do you list files, check and change the current directory, and create directories and files? **[asked x5 in your bank]**

62. How do you view file contents, and copy, move and delete files? **[asked x2 in your bank]**